

Skills turn long research threads into durable paper artifacts

Discord context → local evidence → skill-specific artifact → Thread Canvas → paper/deck plan.

Nonlinear SAE

Value Action

Model Merging

Branch Specialization

Core loop

Discord thread → local artifacts → skill-guided output → curated Thread Canvas.

Artifact rule

Show the exact PDF/HTML/PPTX that supports the next decision.

Claim rule

Keep evidence classes and overclaim boundaries visible.

Near-real research questions

Nonlinear SAE, Value Action, Model Merging, and Branch Specialization map to concrete paper/deck nodes.

Nonlinear SAE

Do residual/nonlinear encoders improve EV-L0 tradeoffs by support selection?

Value Action

Can sparse features track value/action commitments before the answer?

Model Merging

Can sparse-basis interventions repair refusal while preserving capabilities?

Branch Specialization

Are head role/family affinities real before stronger modularity claims?

Matched comparison before headline claims

Matched EV-L0 beat/loss maps come before global ranking and semantic interpretability claims.

eligible runs → dead-neuron gate
matched pairs → EV/L0 beat-loss
global ranks → EV + reverse-L0 shortlist
semantic lane → DAME/autointerp

Safe claim

Nonlinear encoders can improve sparse support selection under matched comparisons.

Avoid

Do not claim universal interpretability without semantic/redundancy gates.

S-lane presence vs answer-gap lane

S1/S2/S3 define discovery/presence; S4 captures pre-answer contribution gaps; V1/V2 validate causal decisiveness.

S1/S2/S3

presence + intensity screens

S4

pre-answer contribution gap

V1/V2

causal decisiveness validation

Promote the small human-facing artifact set

Paper-preview HTML, deck PDF/PPTX, contact sheet, and ledger are promoted; raw scratch files are not.

Keep

paper preview HTML

Keep

deck PDF/PPTX

Keep

contact sheet + ledger

Every result gets a manuscript location

Each result receives a main-paper section, appendix location, evidence class, claim boundary, and next gate.

Result	Main	Appx	Gate
ResNet vs Linear EV-L0	RQ1	D	show wins and losses
DAME semantic diagnostics	RQ2/Appx	E/F	dominance-aware validation
ConsVal row-level tiers	RQ2	C/D	pooled vs order-robust labels

Decks are rendered artifacts, not outlines

PPTX is rendered to PDF, slide images/contact sheet are inspected, and dense pages are fixed before delivery.

PDF

Render before delivery.

Contact sheet

Inspect for clipping and overflow.

Backing files

Link CSV/HTML/source.

The skill library is a research operating system

Reusable skills encode source-of-truth files, artifact curation, paper placement, deck QA, and safe claims.

show work

show gates

show artifacts

show next step

Public-safe showcase

This repo shows the workflow shape and standards without exposing raw private data or secrets.